



Seasonal Adjustment and Promotion Impact Analysis

Final Client Report

Prepared for
Farmer Boys

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This document outlines the methodology, findings and recommendations for the Farmer Boys

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Executive Summary

Farmer Boys, a regional fast-casual restaurant chain operating 104 locations across California, Arizona, and Nevada, engaged the team to answer a deceptively simple question: behind the weekly ups and downs of sales, is the business actually growing or quietly declining? Seasonal swings—holidays, spring promotions, and the fiscal calendar—were large enough to obscure the impact of operational decisions and promotions, making it difficult to read true performance or to plan with confidence.

To resolve this, the team built two linked analytical engines from four-plus years of weekly store-level sales and transaction data. The first is a **seasonal-adjustment model** - a 54-variable regression that strips recurring calendar effects out of the data to reveal an underlying “baseline.” The second is a **multi-model forecasting system** that blends statistical and machine-learning methods into an ensemble, validated through 40 rolling backtests, to project the next quarter and the full year ahead for both revenue and customer traffic.

The central finding is a traffic problem that revenue had been masking. Once seasonality is removed, customer transactions have declined every year, falling roughly 18% from FY2023 to FY2026, while sales held up far longer because customers were spending more per visit. By FY2026 both measures sat below their multi-year norm. In short, the decline in visits began years before it became visible in the top line — and it is the more urgent issue.

Looking forward, the models project continued, traffic-led softness in FY2027. The forecast points to a low-to-mid single-digit decline in full-year sales and a steeper decline in transactions versus the chain’s recent three-year benchmark, with average spend per visit remaining comparatively stable. Both forecasting engines validated with high accuracy (sales error of roughly 1.3% and transaction error of roughly 1.4% in out-of-sample testing), giving the projections a sound, tested basis for planning.

The team’s top recommendations are to: (1) treat declining traffic — not headline sales — as the primary business priority; (2) evaluate every promotion against the seasonally adjusted baseline so its true incremental lift is measured rather than mistaken for a seasonal peak; (3) use the ensemble forecast as the basis for quarterly operational planning and annual budgeting; and (4) refresh the projections each quarter using the self-service tools delivered with this engagement. Those tools — a baseline model, a forecasting system, an interactive dashboard, and an Excel workbook — ensure the analytical value of this project persists well beyond it.

1. Context and Background

Farmer Boys is a regional fast-casual restaurant chain known for its breakfast, burgers, and farm-fresh positioning. The company operates 104 locations across California, Arizona, and Nevada and competes in a crowded, price-sensitive segment where customer traffic and per-visit spend are the two fundamental levers of revenue. Like most multi-unit restaurant operators, Farmer Boys experiences pronounced weekly and seasonal demand cycles: spring weeks and major sporting and holiday events drive volume up, while late-year holidays such as Thanksgiving and Christmas pull it sharply down.

Management tracks weekly sales and transactions closely, but those raw figures move so much from week to week that they are difficult to interpret. A strong week may simply reflect a favorable point in the calendar rather than a genuine improvement in operations, and a soft week may reflect a holiday rather than a real loss of demand. This made it hard to answer routine but important questions: Did a promotion actually work? Is the business healthier this year than last? What should next quarter's sales and staffing plan assume?

The team worked with multi-year, store-level weekly data covering FY2020 and the continuous FY2023–FY2026 period, drawn from a sample of 85 stores. Farmer Boys follows a February–January fiscal year, and a fiscal week occasionally falls in a 53-week year, both of which had to be handled carefully so that comparisons across years remained valid.

An initial look at the raw figures showed why interpretation was so difficult. Total annual sales grew strongly in the earlier part of the window — up roughly 20% between FY2020 and FY2023 — then peaked near \$204.6 million in FY2024 before easing to about \$193.7 million in FY2026, a decline of roughly 5% from the peak. Customer transactions, however, did not follow the same arc: annual transaction counts fell in every year of the recent window, from about 11.4 million in FY2023 to about 9.3 million in FY2026. The raw data also displayed a pronounced and stable seasonal shape — the spring quarter is consistently the strongest, fiscal weeks 14 through 18 are the highest-volume weeks of the year, and the weeks around Thanksgiving and Christmas are the weakest. These patterns confirmed that any honest read of performance would first have to account for the calendar. With that data and context in place, the engagement set out to separate what is seasonal and expected from what is real and actionable.

2. Problem Statement

Seasonal fluctuations were obscuring Farmer Boys' true operating performance. The core business problem was that recurring calendar effects in weekly sales and transactions were large enough to hide the underlying trend in the business and to distort the apparent impact of promotions and operational changes. Without a way to remove that seasonal “noise,” the company could neither measure performance reliably nor plan against it.

The engagement was scoped around two connected objectives:

1. **Build a seasonally adjusted baseline.** Strip recurring weekly and holiday effects out of sales and transactions so that the relative strength of underlying performance — and the genuine, incremental impact of promotional and operational activity — can be seen and measured.
2. **Forecast sales and transactions.** Produce reliable forward projections of revenue and customer traffic that account for seasonality, recent trend, and evolving business patterns, at both a 13-week (next-quarter) and a full-year horizon.

This problem matters because nearly every near-term operating decision depends on a clear read of demand. Labor scheduling, inventory ordering, promotional calendars, and annual budgets are all built on expectations of weekly sales and traffic. If those expectations are anchored to numbers distorted by seasonality, the resulting plans are systematically wrong. A successful outcome was therefore defined as a validated, repeatable capability — not a one-time analysis — that lets Farmer Boys (a) read its true performance net of seasonality and (b) forecast the coming quarter and year with quantified, tested accuracy.

3. Analytical and Technical Approach and Results

This section walks through the full analytical journey in four parts: the overall approach and why it was chosen, the data and how it was prepared, the seasonal-adjustment work and what it revealed, and the forecasting work and its projections. It closes with validation and limitations. Detailed model specifications, coefficient tables, and accuracy statistics are provided in the appendices and referenced throughout.

3.1 Approach

The team chose a two-stage design that mirrors the two business objectives. The first stage is **explanatory**: a transparent regression that decomposes each week's sales and transactions into a recurring seasonal component, a long-run trend, and a remaining baseline. The second stage is **predictive**: an ensemble of forecasting models that learns from history to project the future. The two stages are deliberately separate. A single “black-box” model could have produced forecasts, but it would not have given management an interpretable view of why the business moves as it does. Pairing an interpretable decomposition with a high-accuracy forecasting ensemble delivers both understanding and prediction.

For the seasonal adjustment, ordinary least squares (OLS) regression was selected because it is transparent, produces directly interpretable dollar-and-visit effects for each week and holiday, and allows every effect to be tested for statistical significance. For forecasting, no single method reliably dominates restaurant demand data, so the team tested a broad library of statistical and machine-learning models and combined the strongest performers into a weighted ensemble. Because customer traffic behaves differently from revenue — it is noisier and more driven by momentum and behavioral shifts than by the calendar — transactions were given their own dedicated model rather than being derived mechanically from sales.

3.2 Data

The analysis drew on Farmer Boys' own weekly operating data: store-level sales revenue and transaction counts for a sample of 85 stores, spanning FY2020 and the continuous FY2023–FY2026 period (with a 13-week view into early FY2027 used for context). The data arrived in a “wide” Excel layout — one sheet per metric, with stores down the rows and year-week combinations across the columns — which is convenient for review but unusable for modeling.

A reusable conversion routine reshaped this into a clean “long” format with one row per fiscal year and week and four columns: year, week, total sales, and total transactions. The routine reads both the sales and transaction sheets, aligns them on the fiscal year and week labels, sums across all stores to the chain level, fills isolated gaps with zero, and writes a single analysis-ready file. Several fiscal-calendar details were handled explicitly: the February–January fiscal year, the occasional 53-week year (whose extra week is mapped onto week 52 so seasonal comparisons stay aligned), and the fact that Thanksgiving shifts between fiscal weeks 47 and 48 from year to year. Standardizing week labels in this way was essential for valid year-over-year comparison and is the foundation on which every later result rests. The conversion logic is delivered to Farmer Boys as documented code (see Appendix E).

Because the FY2020 data sat apart from the later years and the modeling work required an unbroken weekly history, the continuous FY2023–FY2026 span — roughly 208 consecutive weeks — served as the core training window, with the most recent period reserved for out-of-sample validation. Data quality was checked at this stage: weekly totals were reconciled against the source workbook, week counts per fiscal year were confirmed (52 in standard years, 53 in the long year), and no material gaps remained after the alignment step. This produced a single, trustworthy series for each metric on which both the seasonal-adjustment and forecasting models were built.

3.3 Seasonal Adjustment: Methodology and Findings

To isolate underlying performance, the team fit a 54-variable OLS regression to weekly chain-level data for both sales and transactions. The model includes 52 fiscal-week indicator variables that capture the recurring pattern of each week in the year, a dedicated Thanksgiving flag (because Thanksgiving moves between weeks 47 and 48 and would otherwise contaminate those weeks' estimates), and a single linear fiscal-year trend term that captures long-run business drift. The seasonally adjusted baseline for each week is then defined simply as actual performance minus the estimated seasonal effect, leaving the trend and the genuine, non-seasonal movement intact.

The models fit the data well and every effect was statistically significant at the 5% level. The sales model explained 83% of weekly variation ($R^2 = 0.83$) and the transaction model 94% ($R^2 = 0.94$), confirming that a large share of week-to-week movement is calendar-driven and can be reliably removed. The estimated effects are intuitive: spring weeks 14–16 and the Super Bowl and pre-Easter weeks add the most volume, while Thanksgiving week is by far the largest drag, reducing chain sales by roughly \$451,000 and transactions by about 24,000 in that week alone. Full coefficient tables appear in Appendix A.

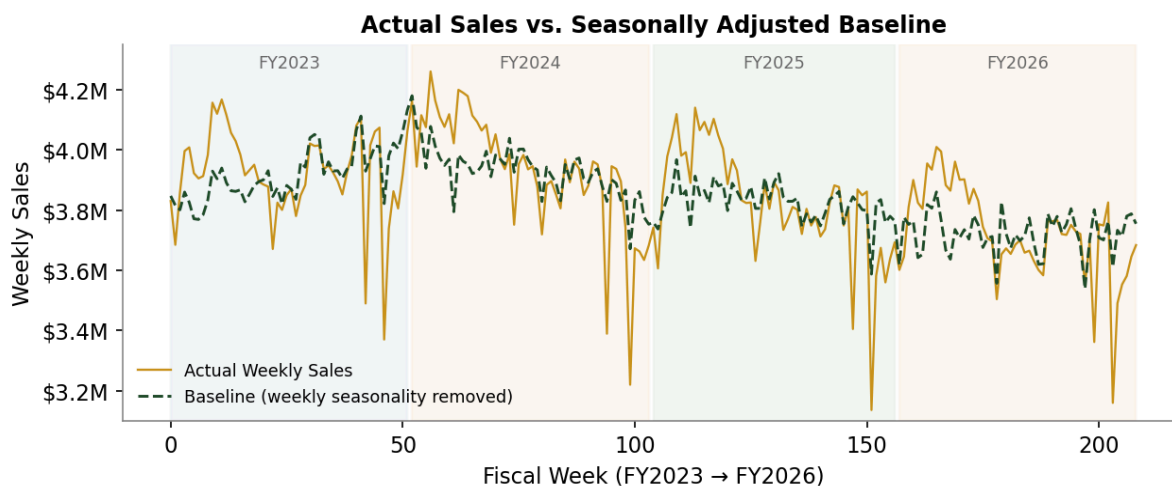


Figure 1. Actual weekly sales (gold) versus the seasonally adjusted baseline (green). Removing recurring calendar effects reveals an underlying trend that drifts downward in the most recent years, even as seasonal peaks continue to mask it in the raw series.

The most important insight emerged when the adjusted baselines were indexed across years. Setting a weighted three-year norm to 100, the team compared each year's seasonally adjusted average for sales and for transactions (Table 1). The two series tell strikingly different stories. Seasonally adjusted sales held above the norm through FY2025 and only dipped below it in FY2026. Seasonally adjusted transactions, by contrast, fell every single year — from an index of 116.6 in FY2023 to 95.4 in FY2026, a decline of roughly 18%.

Fiscal Year	Sales SA Avg/Week	Sales Index	Txn SA Avg/Week	Transaction Index
FY2023	\$3.917M	102.6	218.5K	116.6
FY2024	\$3.929M	103.0	203.8K	108.7
FY2025	\$3.898M	102.2	190.8K	101.8
FY2026	\$3.721M	97.5	178.9K	95.4

Table 1. Seasonally adjusted performance index by fiscal year (base = 100, the weighted average of FY2024–FY2026). Sales stayed near or above the norm until FY2026; transactions declined every year.

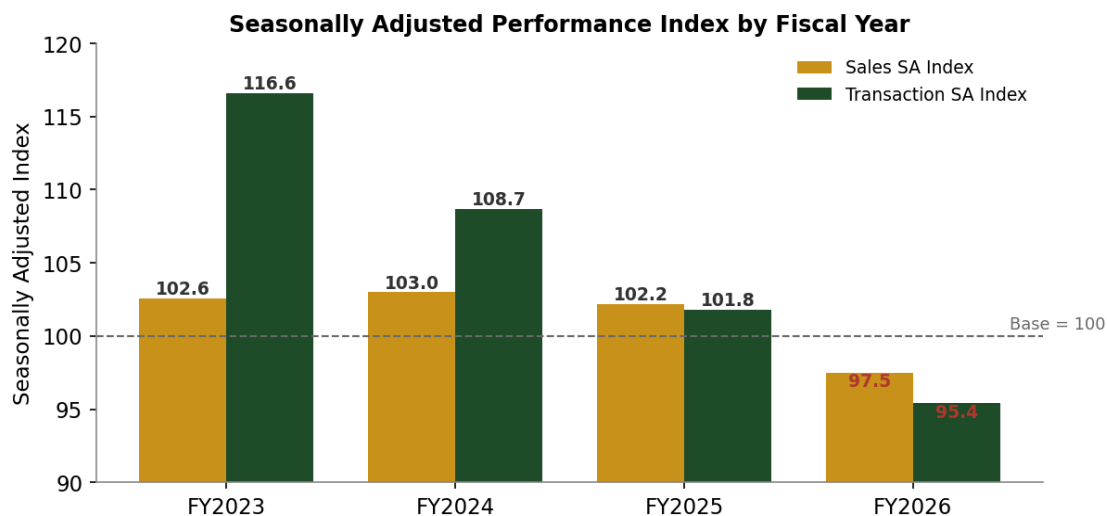


Figure 2. The same indices shown side by side. The widening gap between sales (gold) and transactions (green) is the signature of fewer customers spending more per visit.

Read together, these results reveal that a traffic decline preceded the sales decline by roughly three years and was masked by rising spend per visit. As long as the average customer kept spending more, falling visit counts did not show up in revenue. That cushion thinned over time, and by FY2026 both measures had slipped below their multi-year norm. The seasonal-adjustment work thus reframes the business question: the headline issue is not a recent dip in sales but a sustained erosion of customer traffic that the chain’s rising average ticket had been quietly offsetting.

3.4 Forecasting: Methodology

With the underlying trend exposed, the second stage projects it forward. The forecasting process followed five disciplined steps for sales and an extended six-step process for transactions: engineer predictive features, test many candidate models independently, validate them through rolling backtests, optimize the blend with a grid search, and generate the final 13-week and full-year forecasts (with an additional hybrid-blending step for transactions, described below).

Feature engineering

Each model was supplied with features designed to encode the signals that drive demand: lagged values at 1, 2, 4, 13, and 52 weeks (with the 52-week, same-week-last-year lag the single strongest predictor); 4- and 13-week rolling averages to smooth noise; year-over-year change signals; Fourier terms to represent smooth seasonal waves; a trend index; and holiday flags. Because traffic is noisier and more momentum-driven, the transaction models added linear and quadratic trend terms, short-horizon momentum features (week-over-week differences and percentage changes at 1, 4, and 13 weeks), and a wider set of lags and rolling windows. The full feature dictionary is in Appendix C.

Model library and ensemble

Rather than commit to one method, the team evaluated fourteen candidate models — nine statistical (including ETS, ARIMA/SARIMA, Theta, Fourier OLS, Week-Flags OLS, Ridge, and a detrend-then-ETS

approach) and five machine-learning (LightGBM in two configurations, XGBoost, Random Forest, and Extra Trees). Each captures a different facet of the data: week-flag regressions nail exact fiscal-week and holiday behavior, ETS methods track stable recurring seasonality and damp unrealistic long-run projections, and gradient-boosted trees capture nonlinear shifts and interactions. No single model excelled on every dimension, so the strongest performers were combined into a weighted ensemble that balances their individual strengths. The selected models and their ensemble weights are listed in Appendix B.

Rolling backtests and weight optimization

To estimate honest, real-world accuracy, the team used a rolling-origin backtest that mimics how the tool will actually be used. Each iteration trains on a 156-week window (three fiscal years), forecasts the next 13 weeks, then slides forward one week and repeats — 40 windows in total. This produces out-of-sample errors across many different starting points rather than a single, potentially lucky, holdout. A grid search over thousands of weight combinations then tuned how much each model contributes to the ensemble, selecting the blend that minimized error across all backtest windows.

Hybrid transaction model

Transactions received one additional safeguard. The final transaction forecast is a hybrid: 80% comes from the dedicated five-model transaction ensemble (which captures traffic-specific seasonality and momentum), and 20% comes from a sales-derived estimate — the sales forecast divided by a predicted weekly average ticket, where the ticket itself is forecast as a weighted blend of the last three years (50% most recent, 30% two years prior, 20% three years prior). Anchoring 20% of the traffic forecast to the sales forecast keeps the two projections internally consistent and prevents the noisier traffic model from drifting into unrealistic territory, while still letting it express genuine behavioral patterns.

3.5 Forecasting: Results

The validated ensembles were used to generate FY2027 forecasts at two horizons — the next 13 weeks (the FY2027 Q2 operating quarter) and the full 52-week year — for both sales and transactions. Each forecast is reported against a three-year weighted benchmark that represents the chain's recent normal level of business, so the comparison isolates how FY2027 is expected to differ from recent performance. Table 2 summarizes the headline results.

Metric	13-Week (Q2)	vs Benchmark	Full Year	vs Benchmark
Sales	~\$49.0M	-4.5%	~\$189.4M	-4.1%
Transactions	~2.37M	-7.2%	~9.28M	-4.3%
Avg. ticket	~\$20.65	roughly flat	~\$20.42	roughly flat

Table 2. FY2027 forecast summary versus the three-year weighted benchmark. Declines are concentrated in transactions; average spend per visit is projected to remain comparatively stable.

The forecast extends the same story the baseline revealed: continued, traffic-led softness. Full-year sales are projected at roughly \$189 million, a low-single-digit decline against the three-year benchmark,

while transactions are projected to fall more sharply. Crucially, the projected average ticket holds near \$20, confirming that the pressure on the business comes from fewer visits rather than from customers spending less. The near-term (13-week) picture is somewhat softer than the full-year average, consistent with the seasonal profile of the FY2027 Q2 window. Figures 3 and 4 place these forecasts in the context of recent actuals and the benchmark.

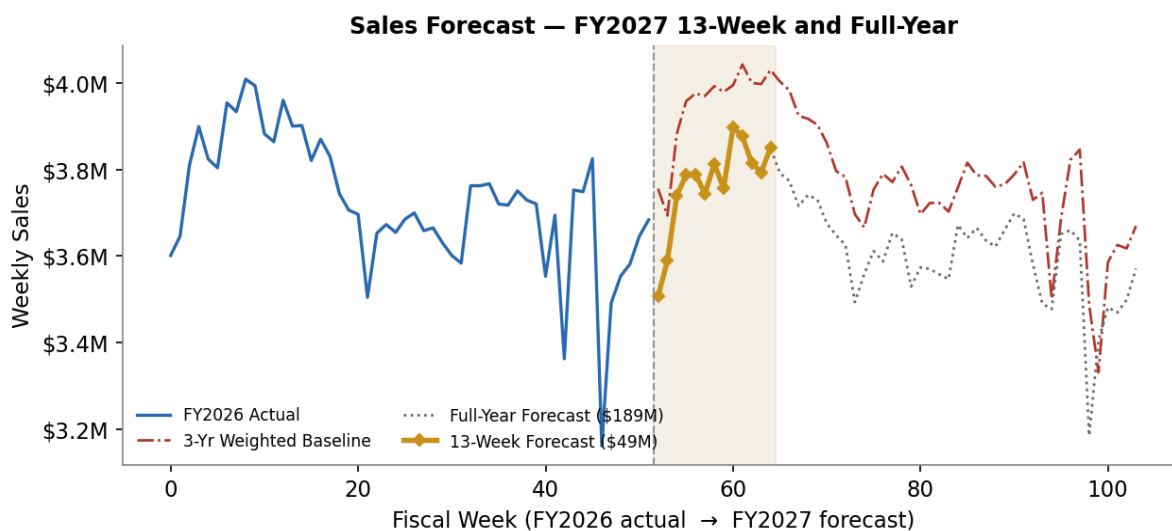


Figure 3. FY2027 sales forecast. The 13-week forecast (gold) and full-year forecast (dotted) sit below the three-year weighted benchmark (red), extending the downward baseline trend into the coming year.

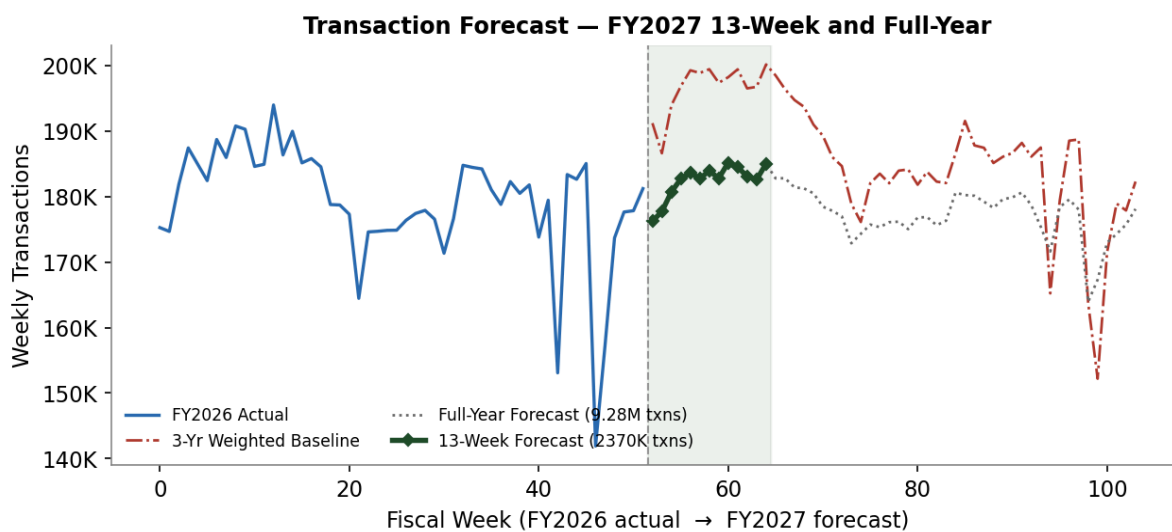


Figure 4. FY2027 transaction forecast. Projected visits remain below the benchmark across the year, the clearest single indicator of the traffic challenge facing the chain.

3.6 Validation and Limitations

The forecasts were validated on out-of-sample data, not merely fit to history. Across the 40 rolling backtest windows, the sales ensemble achieved a mean absolute percentage error (MAPE) of about 1.3% with an R^2 of roughly 0.81, and the hybrid transaction ensemble achieved a MAPE of about 1.4% with an

R^2 of roughly 0.81. In both cases the ensemble outperformed every individual model, confirming the value of blending. These error levels — on the order of 1–2% of weekly volume — are well within the range needed for operational and budgeting use. Detailed accuracy statistics for the ensemble and each component model are provided in Appendix D.

Several assumptions and limitations should inform how the results are used:

- **Chain-level scope.** Models were built at the chain level on a sample of 85 stores. They describe system-wide patterns and are not a substitute for store-level forecasts, where local effects would matter more.
- **History-based assumptions.** The models assume that the relationships seen in the past continue to hold. A structural change — a major menu overhaul, pricing reset, new-unit growth, or macroeconomic shock — would not be anticipated until it appears in the data.
- **No external drivers.** Weather, gas prices, local promotions, and competitor activity were not included as inputs. These were identified as promising future variables but fell outside the current scope.
- **Benchmark sensitivity.** Forecast-versus-benchmark percentages depend on the three-year weighted benchmark used as the comparison; the absolute forecasts are the more stable quantities. As new weeks of actual data are loaded, both the forecasts and the benchmark refresh, so the point estimates above should be treated as a snapshot rather than a fixed figure.

None of these limitations undermines the central conclusions, which are robust across every view of the data: the seasonally adjusted trend is downward, the decline is driven by traffic rather than spend, and that pattern is projected to continue absent intervention.

4. Key Takeaways and Recommendations

The analysis translates into a small number of clear insights and a focused set of actions.

4.1 Key Takeaways

1. **Raw sales overstate the health of the business.** Roughly four-fifths of weekly variation in sales and nine-tenths in transactions is calendar-driven. Reading raw figures without adjustment leads to misjudging both performance and the effect of decisions; the seasonally adjusted baseline is the correct lens.
2. **Farmer Boys has a traffic problem, not (yet) a spending problem.** Seasonally adjusted transactions fell about 18% from FY2023 to FY2026, while average spend per visit rose enough to mask it in revenue until FY2026. The most urgent metric to manage is visits, not the top line.
3. **The traffic decline is sustained and is projected to continue.** Both the historical baseline and the FY2027 forecast point the same direction: continued, traffic-led softness against the chain's recent norm, with spend per visit holding roughly flat.

4. **An ensemble forecast is materially more reliable than any single model.** Blending statistical and machine-learning models, validated through rolling backtests, delivered out-of-sample errors near 1–2% — a tested, trustworthy basis for planning.
5. **The capability is now self-service.** The value of this work persists because it was delivered as reusable tools, not a one-time report. Farmer Boys can regenerate baselines and forecasts in-house as new data arrives.

4.2 Recommendations

1. **Make customer traffic the primary business priority.** Reframe internal performance reviews and goals around seasonally adjusted transactions, and direct marketing and operational initiatives at reversing the multi-year decline in visits. The data shows this is the root issue that rising average tickets have been masking; treating it as the headline metric is the single most important change recommended.
2. **Evaluate every promotion against the seasonally adjusted baseline.** Before crediting a promotion with a strong week, compare results to the adjusted baseline for that week. This separates genuine incremental lift from a favorable point in the calendar (a spring week will look strong regardless), allowing marketing spend to be allocated to what truly works.
3. **Adopt the ensemble forecast for quarterly planning and annual budgeting.** Use the 13-week forecast to inform near-term labor, inventory, and promotional decisions, and the full-year forecast as the analytical anchor for the annual budget. The validated ~1–2% accuracy supports using these projections directly in operational planning.
4. **Refresh the forecast each quarter with updated data.** Re-run the self-service tool quarterly as new weeks of sales and transactions become available. This keeps projections current, lets the models absorb any change in trend quickly, and turns forecasting into a routine management rhythm rather than a periodic project.

Expected impact, sequencing, and risks. The immediate benefit is decision quality: management gains an undistorted read of performance and a tested forecast for planning, reducing the risk of over- or under-resourcing driven by seasonal noise. The recommended sequence is to adopt the seasonally adjusted baseline as the standard performance lens first (it requires no new data), then institutionalize the quarterly forecast cycle, and in parallel launch traffic-focused initiatives whose results are now measurable against the baseline. The principal risk to monitor is structural change: because the models learn from history, management should watch for the forecast and the benchmark diverging from actuals over several quarters, which would signal that a new pattern has taken hold and that the models warrant re-examination. A natural next step, should Farmer Boys wish to extend the work, is to incorporate external drivers — weather, gas prices, local events — and to develop store-level versions of the models.

Appendix A. Seasonal Adjustment Model Specification

The seasonal-adjustment model is a 54-variable ordinary least squares regression estimated separately for chain-level weekly sales and transactions:

- 52 fiscal-week indicator (dummy) variables capturing the recurring level of each week of the fiscal year. An extra week in a 53-week fiscal year is mapped onto week 52.
- 1 Thanksgiving indicator, which floats between fiscal weeks 47 and 48 across years and therefore requires its own flag.
- 1 linear fiscal-year trend index capturing long-run business drift.

Baseline definition: Seasonally adjusted baseline = actual value – estimated seasonal effect. All effects were statistically significant at $p < 0.05$. Sales $R^2 = 0.83$; Transactions $R^2 = 0.94$. Table A1 lists the largest estimated weekly effects.

Calendar event (fiscal week)	Sales effect (\$/week)	Transaction effect (visits/week)
Super Bowl (Wk 5)	+\$127,706	+10,295
Pre-Easter (Wk 13)	+\$145,432	+8,567
Spring (Wk 14)	+\$208,634	+13,355
Spring (Wk 15)	+\$183,128	+10,772
Spring (Wk 16)	+\$133,617	+9,967
Thanksgiving	-\$451,134	-23,727

Table A1. Largest seasonal effects (chain level), relative to an average week. Thanksgiving is the single largest negative driver for both metrics.

Appendix B. Forecasting Model Inventory and Ensemble Weights

Fourteen candidate models were tested (nine statistical: ETS Standard, ETS Damped, ARIMA, SARIMA, Theta, Fourier OLS, Week-Flags OLS, Ridge Regression, Detrend+ETS; five machine-learning: LightGBM Standard, LightGBM Ultra, XGBoost, Random Forest, Extra Trees). The best performers were combined into a weighted ensemble for each metric. Weights and per-model backtest accuracy are shown below.

Table B1. Sales ensemble

Model	Weight	R^2	MAPE
Week-Flags OLS	8%	0.31	2.61%
ETS Standard	31%	0.69	1.69%
ETS Damped	23%	0.69	1.68%
LightGBM Standard	38%	0.57	1.57%
Ensemble	—	0.81	1.29%

Table B2. Transaction ensemble (independent component of the hybrid)

Model	Weight	R ²	MAPE
LightGBM Ultra	38%	0.78	1.65%
LightGBM Standard	25%	0.75	1.71%
ETS Damped	19%	0.48	2.78%
Fourier OLS	12%	-0.20	4.39%
Week-Flags OLS	6%	-0.29	4.54%
Ensemble	—	0.82	1.44%

Final transaction forecast = 80% × independent ensemble + 20% × (sales forecast ÷ predicted weekly average ticket). The predicted ticket is a weighted blend of the trailing three years (50% / 30% / 20%, most-recent-weighted).

Appendix C. Feature Dictionary

Sales features

- Lag comparisons: lag_1, lag_2, lag_4 (recent-week momentum); lag_13 (same quarter, prior period); lag_52 (same week last year — strongest predictor).
- Rolling averages: rm4 and rm13 (4- and 13-week means) to smooth weekly noise.
- Year-over-year change signals; Fourier terms for smooth seasonal waves; a trend index; and holiday flags (Thanksgiving floating between weeks 47–48).

Additional transaction features

- Trend: linear and quadratic terms to capture accelerating or slowing traffic.
- Momentum: week-over-week differences and percentage changes at 1, 4, and 13 weeks.
- Wider lag set (1, 7, 13, 26, 52 weeks) and rolling means (4, 7, 13, 26 weeks) to capture short-term and year-over-year momentum and to stabilize noisy movement.

Appendix D. Backtest Accuracy Detail

Accuracy was measured out-of-sample across 40 rolling 13-week windows (156-week training window, sliding forward one week at a time). Ensemble results:

Metric	R ²	MAE	MAPE	RMSE (%)
Sales ensemble	0.81	~\$47.4K	1.29%	1.63%
Transaction ensemble	0.81	~2,508	1.44%	1.84%

Table D1. Out-of-sample ensemble accuracy. MAE is mean absolute error; MAPE is mean absolute percentage error; RMSE (%) is root-mean-square error as a share of the mean. The ensemble outperformed every individual model on both metrics.

Model-output visualization. Figures D1 and D2 reproduce the team’s full diagnostic chart: for each metric, the actual series against the seasonally adjusted baseline, the 13-week backtest of forecast versus actual, and the FY2027 forecast against the three-year weighted benchmark.



Figure D1. Sales — baseline, backtest, and forecast diagnostics.



Figure D2. Transactions — baseline, backtest, and forecast diagnostics.

Appendix E. Self-Service Tools and Reproducibility

The engagement's lasting value is a set of four self-service deliverables that let Farmer Boys regenerate every result in-house:

- **Baseline Performance Model** — the seasonal-adjustment regression that produces adjusted sales and transaction baselines.

- **Robust Forecasting System** — the validated ensemble pipeline for 13-week and full-year forecasts.
- **Dynamic Dashboard** — an interactive web app for exploring baselines and forecasts against the weighted benchmark.
- **Excel Workbook** — a five-tab output workbook: Seasonally Adjusted Sales, Seasonally Adjusted Transactions, Sales Forecast, Transactions Forecast, and Seasonally Adjusted Index.

Input data format

The tools accept a comma-delimited CSV in long format with exactly four columns — year, week, sales, transactions — one row per fiscal year and week, with a minimum of four years of history and no missing weeks (gaps filled with zero). The included conversion routine generates this file automatically from the standard wide Excel export.

Delivered code

- **Fb_convert_engine** — transforms the wide Excel workbook into the required long-format CSV.
- **Fb_Technical_Analysis** — builds the seasonal baselines, performance indices, and forecasts.

Dashboard access is provided separately to the Farmer Boys team. Refreshing the workflow each quarter with updated data keeps all baselines and forecasts current.